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# Beyond Demographics:

## Enhancing Cultural Value Survey Simulation with Multi-Stage Personality-Driven Cognitive Reasoning



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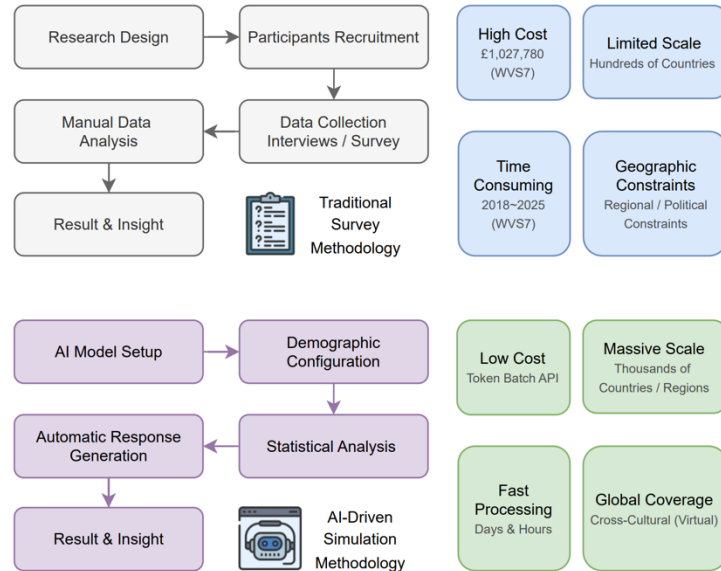
# Why Social Survey Simulation Matters?

## Traditional Challenges:

- High costs and time-consuming human surveys
- Limited scalability for cross-cultural studies
- Resource constraints in social science research

## AI-Driven Solutions:

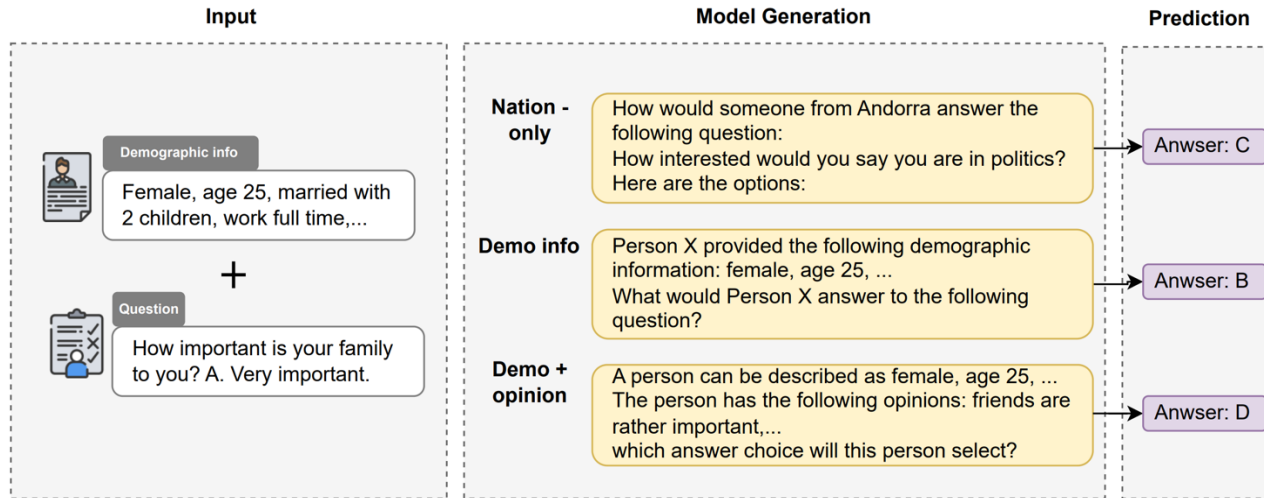
- Growing demand for automated survey simulation
- Need for accurate human behavior modeling
- Potential for large-scale social science insights



[1] <https://www.ukri.org/opportunity/world-values-survey-2025-to-2027/>

[2] <https://catalog.ihsn.org/catalog/12279>

# Current Challenges



## Key Limitations of Existing Methods:

- **Beyond Demographics:** How to go beyond simple demographic features?
- **Psychological Processes:** How to capture human decision-making psychology?
- **Interpretability:** How to provide explainable prediction results?

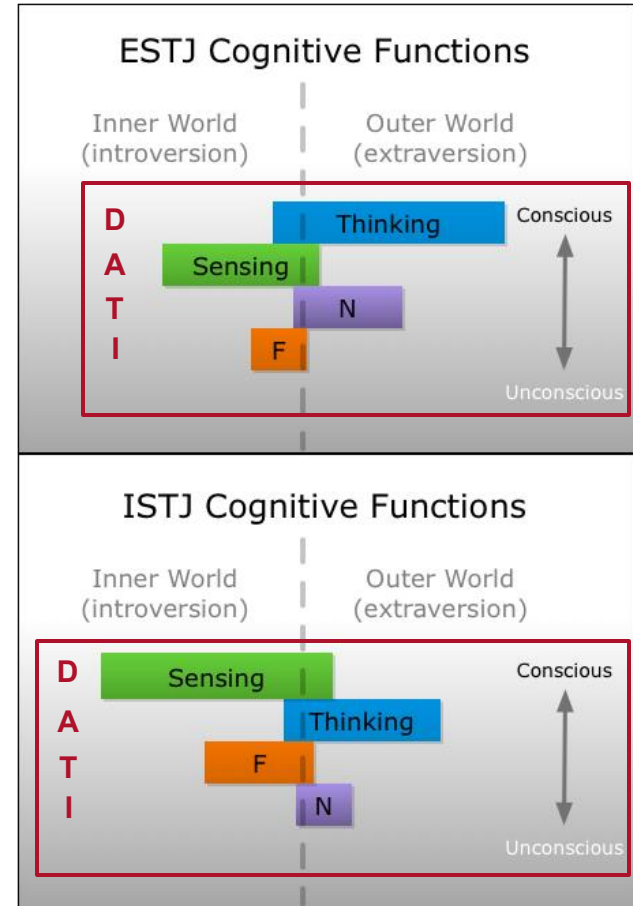
# Key Insight: Psychology Meets AI

## Core Insights:

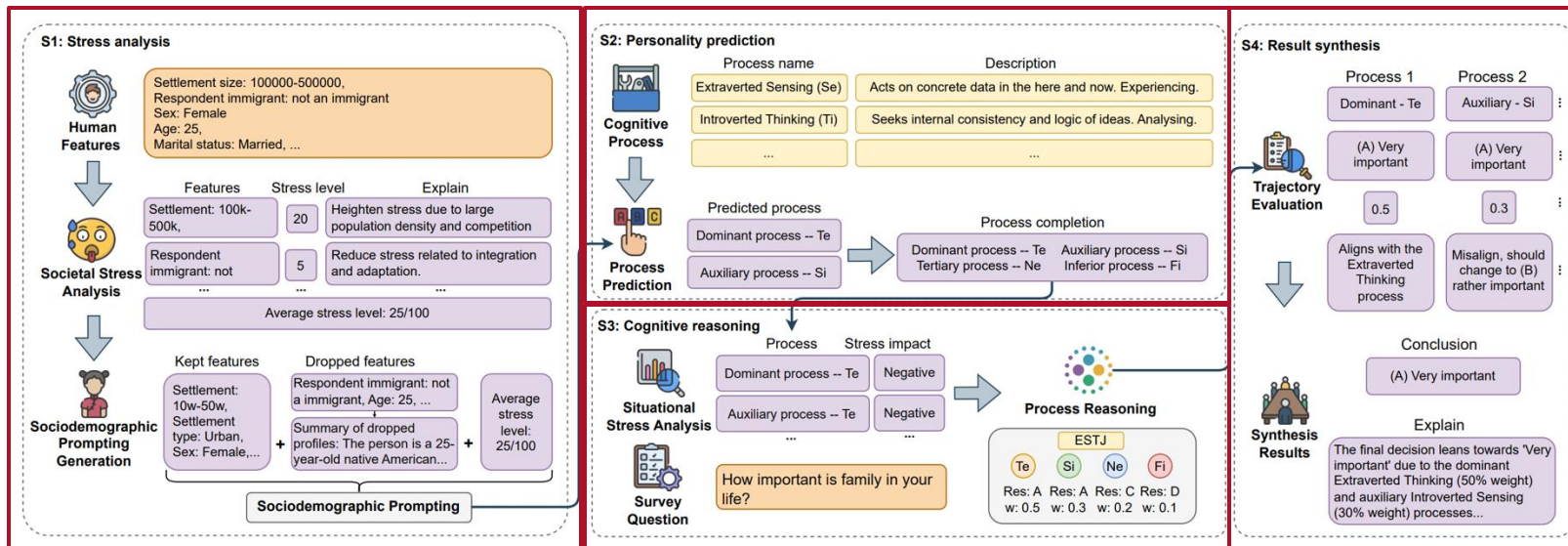
- Human survey responses are influenced by **mental status** (stress levels)
- Different personality types have **distinct cognitive processing patterns**
- MBTI Type Dynamics provides systematic framework

## Research Question

- Can we model psychological theory to enhance AI survey simulation?



# MARK: Multi-stAge Reasoning framework



## Four-Stage Pipeline:

- **Stress Analysis** - Life-situational pressure evaluation
- **Personality Prediction** - MBTI-based cognitive function inference
- **Cognitive Reasoning** - Multi-process weighted simulation
- **Result Synthesis** - Evidence-based decision integration

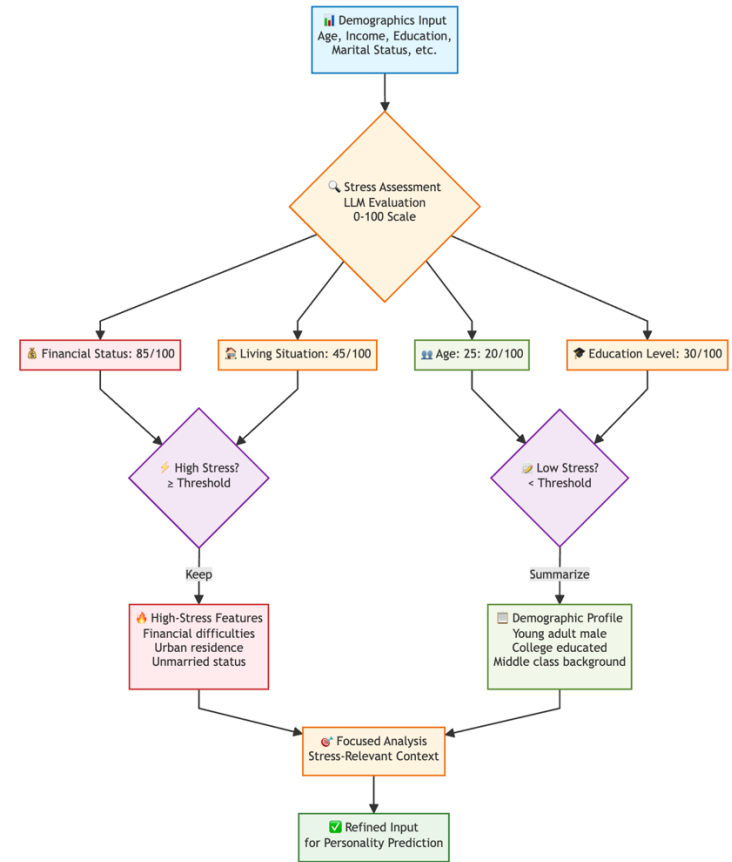
# Stage 1: Life-Situational Stress Analysis

## Stress Evaluation Process:

1. Feature Assessment: Assign stress levels (0-100) to demographic features
2. Automatic Filtering: Preserve high-stress and personality-relevant features
3. Profile Generation: Summarize dropped features into descriptive profiles

## Example

| Feature                | Stress Level | Action    |
|------------------------|--------------|-----------|
| Financial difficulties | 85/100       | Keep      |
| Age: 25                | 20/100       | Summarize |
| Urban residence        | 45/100       | Keep      |



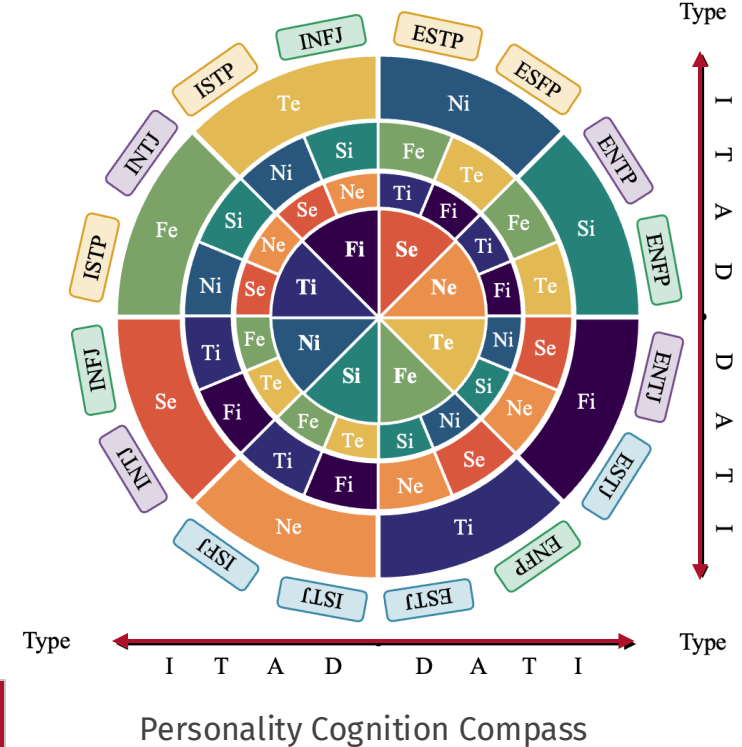
# Stage 2: MBTI-Based Personality Prediction

## Cognitive Function Hierarchy:

- Dominant - Primary, most developed function
- Auxiliary - Supporting, balancing function
- Tertiary - Stress-activated function
- Inferior - Least developed, extreme stress

## Key Innovation

- Use demographics as **environmental context** rather than direct personality mapping



# Stage 3: Weighted Cognitive Process Simulation

## Dual Reasoning Patterns

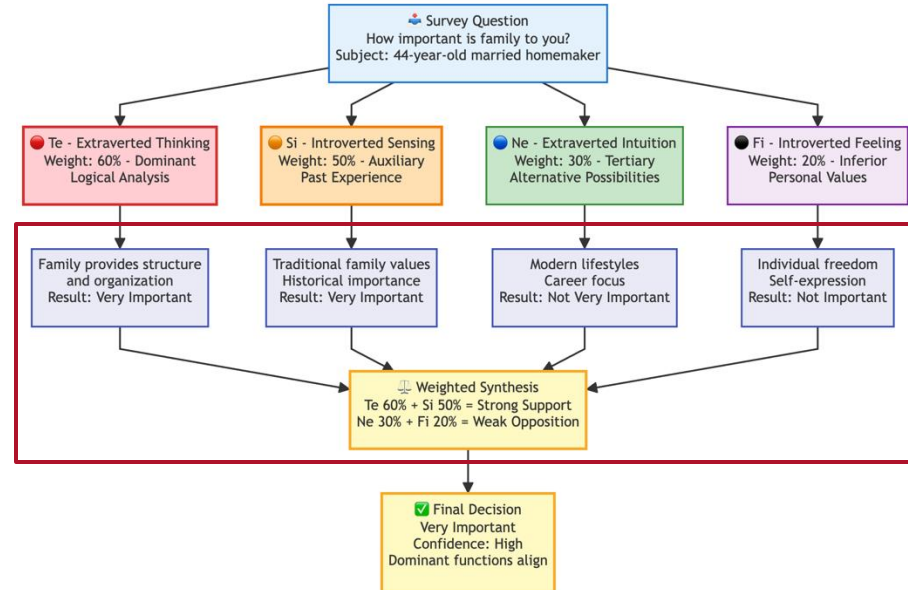
### Normal Functioning

- Low situational stress
- Life-societal stress dominates
- Balanced cognitive processing

### Overused Functioning

- High situational stress
- Trauma-triggered responses
- Cognitive process overload

## Multi-Process Reasoning



# Stage 4: Evidence-Based Decision Synthesis

## Synthesis Process:

1. Trajectory Review: Collect complete reasoning history
2. Consistency Check: Verify alignment with process descriptions
3. Weight Integration: Combine results based on MBTI hierarchy
4. Final Decision: Select most appropriate response

## Decision Example

| Process | Result | Weight | Impact |
|---------|--------|--------|--------|
| Te      | A      | 0.5    | High   |
| Si      | A      | 0.3    | Med    |
| Ne      | B      | 0.2    | Low    |
| Fi      | C      | 0.1    | Low    |
| Final   | A      |        |        |

## Enhanced Interpretability:

Complete reasoning trajectory provides transparent decision-making process

# Experimental Setup

## Dataset

World Values Survey (**7th wave**)

- U.S.: 102,800 QA pairs (20 clusters)
- China: 4,340 QA pairs (20 clusters)
- 30 demographic features
- Group-level clustering approach

## Evaluation Settings

- Global Distribution: Population-level alignment
- Sampled Distribution: Cluster-level performance

## Baseline Methods

- Random selection
- Without demographics
- Demo + Ideology
- Demo + Ideology + Opinion
- Zhao et al. (2024)

## Evaluation Metrics

- ACC: Response accuracy
- 1-JSD: Distribution similarity
- EMD: Earth Mover Distance
- Cohen's Kappa: Group-level agreement

# Main Results: Consistent Improvements

| Model                         | GLM-4-air     |               |            |             | GPT-4o       |               |               |             | Doubao-1.5-pro |               |        |             |
|-------------------------------|---------------|---------------|------------|-------------|--------------|---------------|---------------|-------------|----------------|---------------|--------|-------------|
| Global Distribution           |               |               |            |             |              |               |               |             |                |               |        |             |
| Avg. Metrics                  | 1-JSD ↑       | EMD ↓         | $\kappa$ ↑ | 1-JSD ↑     | EMD ↓        | $\kappa$ ↑    | 1-JSD ↑       | EMD ↓       | $\kappa$ ↑     |               |        |             |
| Random selection <sup>†</sup> | 0.3319        | 0.2848        | 0.0000     | 0.3319      | 0.2848       | 0.0000        | 0.3319        | 0.2848      | 0.0000         |               |        |             |
| Without Demo.                 | 0.4492        | 0.1398        | 0.0125     | 0.4421      | 0.1177       | 0.0138        | 0.3788        | 0.1455      | 0.0112         |               |        |             |
| Durmus et al. (2024)          | 0.2664        | 0.1466        | 0.0142     | 0.4411      | 0.0981       | -0.0071       | 0.3544        | 0.1280      | 0.0116         |               |        |             |
| Liu et al. (2025)             | 0.4371        | 0.0490        | 0.0118     | 0.4620      | 0.0211       | 0.0113        | 0.5258        | 0.0218      | 0.0104         |               |        |             |
| Demo.+Ideo.                   | 0.4336        | 0.1041        | 0.0011     | 0.6404      | 0.0987       | 0.0112        | 0.5520        | 0.1011      | 0.0180         |               |        |             |
| Demo.+Ideo.+Opinion           | 0.5425        | 0.1112        | 0.0180     | 0.6449      | 0.0981       | 0.0097        | 0.5695        | 0.0995      | 0.0198         |               |        |             |
| Zhao et al. (2024a)           | 0.4267        | 0.1280        | 0.0024     | 0.6174      | 0.1113       | 0.0111        | 0.3974        | 0.1265      | 0.0123         |               |        |             |
| MARK (Ours)                   | <b>0.5963</b> | <b>0.1026</b> | 0.0127     | 0.6438      | 0.0985       | <b>0.0128</b> | <b>0.6111</b> | 0.1141      | 0.0123         |               |        |             |
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## Key Achievements

- 10% accuracy improvement over existing baselines
- Consistent performance across GLM-4-air, GPT-4o, Doubao-1.5-pro
- Better distribution alignment with reduced divergence

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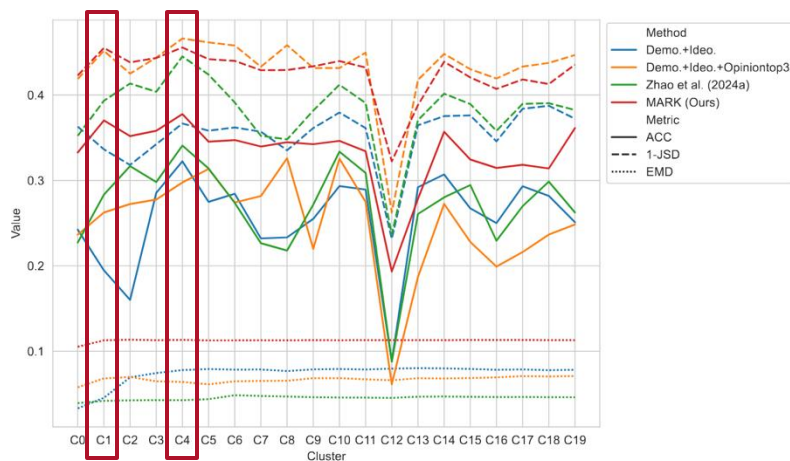
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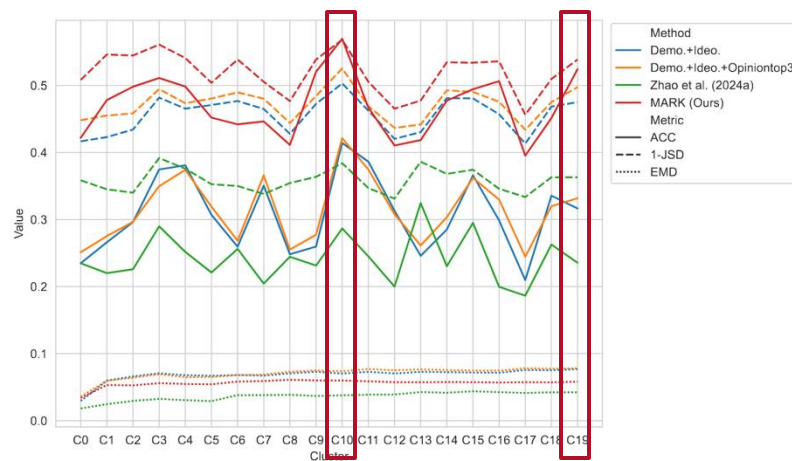
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# Detailed Performance Analysis



(a) Performance of GLM-4-air model.

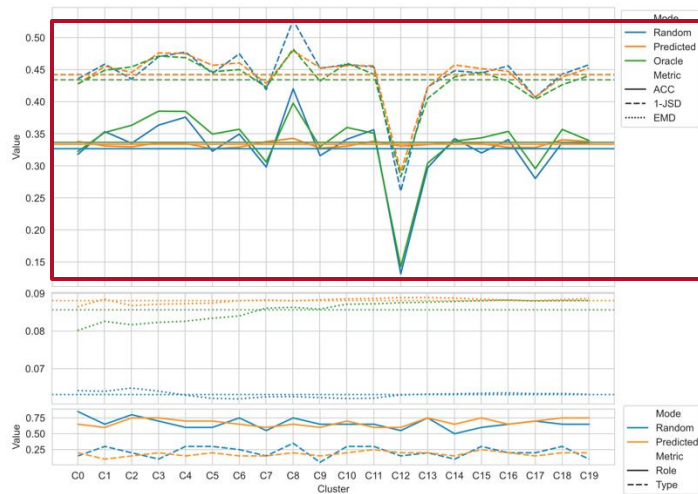


(b) Performance of Doubao-1.5-pro model.

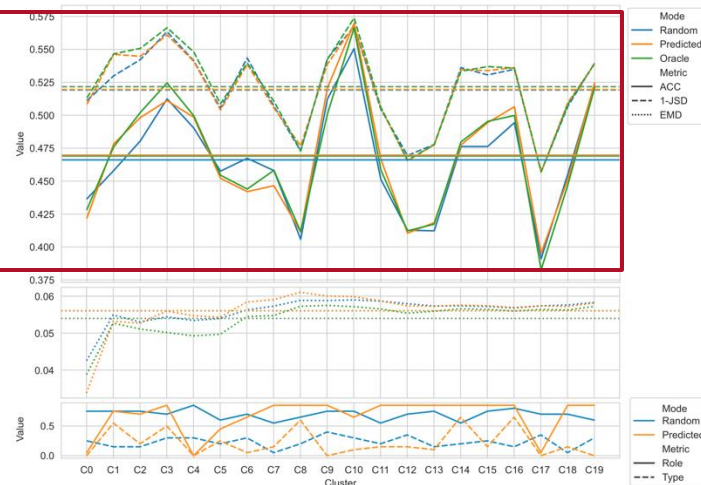
## Cluster-Level Insights

- **GLM-4-air**: Strong performance in married full-time workers (C1: 34.28%, C4: 34.59%)
- **Doubao-1.5-pro**: Excels in skilled private business workers (C10: 56.96%, C19: 52.45%)
- **Differential sensitivity** to demographic backgrounds demonstrates model specialization

# Robustness Analysis



(a) Performance of GLM-4-air model.



(b) Performance of Doubao-1.5-pro model.

## Robustness Finding

MARK maintains stable performance across all personality assignment strategies, demonstrating resilience to personality uncertainty.

## Personality Assignment Strategies

- Predicted: Model-inferred personalities
- Random: Randomly assigned personalities
- Oracle: Response-aware personalities

# Cross-Cultural Generalization

| Method                              | avg_ACC       | avg_1-JSD↑    | avg_EMD↓      |
|-------------------------------------|---------------|---------------|---------------|
| GLM-4-air                           |               |               |               |
| Demo.+Ideo.                         | 25.48%        | 0.4206        | 0.0848        |
| Demo.+Ideo.+Opiniontop3             | 35.21%        | 0.5293        | 0.0861        |
| <a href="#">Zhao et al. (2024a)</a> | 27.07%        | 0.4169        | 0.0932        |
| <b>MARK (Ours)</b>                  | 31.58%        | <b>0.5704</b> | <b>0.0620</b> |
| Doubao-1.5pro                       |               |               |               |
| Demo.+Ideo.                         | 30.35%        | 0.5007        | 0.0810        |
| Demo.+Ideo.+Opiniontop3             | 30.55%        | 0.5128        | 0.0830        |
| <a href="#">Zhao et al. (2024a)</a> | 30.28%        | 0.4407        | 0.0897        |
| <b>MARK (Ours)</b>                  | <b>45.16%</b> | <b>0.6010</b> | 0.0861        |

(a) Result on Chinese prompts.

| Method                              | avg_ACC       | avg_1-JSD↑    | avg_EMD↓      |
|-------------------------------------|---------------|---------------|---------------|
| GLM-4-air                           |               |               |               |
| Demo.+Ideo.                         | 29.47%        | 0.4670        | 0.0877        |
| Demo.+Ideo.+Opiniontop3             | 35.60%        | 0.5339        | 0.0875        |
| <a href="#">Zhao et al. (2024a)</a> | 26.24%        | 0.4054        | 0.0997        |
| <b>MARK (Ours)</b>                  | 31.37%        | <b>0.5793</b> | <b>0.0580</b> |
| Doubao-1.5pro                       |               |               |               |
| Demo.+Ideo.                         | 26.98%        | 0.5039        | 0.0888        |
| Demo.+Ideo.+Opiniontop3             | 27.44%        | 0.5244        | 0.0884        |
| <a href="#">Zhao et al. (2024a)</a> | 32.24%        | 0.4515        | 0.0923        |
| <b>MARK (Ours)</b>                  | <b>44.08%</b> | <b>0.6001</b> | <b>0.0839</b> |

(b) Results on English prompts.

## Cross-Cultural Performance

### Chinese Context:

- Doubao-1.5-pro: 45.16% ACC (Chinese prompts)
- Consistent superiority over baselines
- Strong distributional alignment

**Cultural Adaptability:** MARK successfully generalizes across Eastern and Western cultural contexts

## Language Robustness

- English prompts: 44.08% ACC
- Minimal performance difference
- Cultural adaptability confirmed

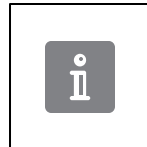
# Ablation Study: Why Four Processes Matter

| Method                    | avg_ACC (%)  | avg_1-JSD↑    | avg_EMD↓      |
|---------------------------|--------------|---------------|---------------|
| Dominant process only     | 16.25        | 0.3126        | 0.1856        |
| Auxiliary process only    | 16.21        | 0.3124        | 0.1856        |
| Tertiary process only     | 16.19        | 0.3116        | 0.1856        |
| Inferior process only     | 15.74        | 0.3074        | 0.1851        |
| <b>MARK (Full system)</b> | <b>33.69</b> | <b>0.4348</b> | <b>0.0887</b> |



## Single Process Limitations

- Dominant only: 16.25% ACC
- Auxiliary only: 16.21% ACC
- Tertiary only: 16.1a9% ACC
- Inferior only: 15.74% ACC



## Multi-Process Integration

- Full MARK System: 33.69% ACC  
- More than **double** the performance of single processes

# Case Study: How MARK Works in Practice



## Subject Profile:

- 44-year-old married homemaker, 2 children,
- urban Georgia,
- Catholic, high school education



## Question:

- How important is family to you?



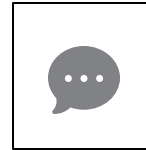
## Stage Outputs:

- **Stress Analysis:** 46.5/100 average
- **Personality:** ISFJ predicted



## Cognitive Reasoning:

- Si (60%): Very important
- Fe (50%): Very important
- Ti (30%): Not very important
- Ne (20%): Not important at all



## Final Decision:

- **Result:** "Very important"
- **Reasoning:** Dominant Si and auxiliary Fe processes both strongly align with family values, overriding minority tertiary and inferior processes.

# Key Contributions & Impact

## Technical Contributions

- **Novel Framework:** First MBTI-based multi-stage reasoning system for survey simulation
- **Performance Gains:** 10% accuracy improvement with cross-cultural generalization
- **Enhanced Interpretability:** Complete reasoning trajectory and decision transparency

## Broader Impact

- **Social Science Research:** Cost-effective alternative to human surveys
- **Cross-Cultural Studies:** Scalable cultural value analysis
- **AI Psychology:** Bridge between psychological theory and computational modeling
- **Interdisciplinary Innovation:** New paradigm for psychology-informed AI

# Limitations & Future Directions

## Current Limitations

- **Personality Dependency:** Requires accurate personality prediction
- **Cultural Coverage:** Limited to US-China contexts
- **Model Constraints:** API-based closed-source models
- **MBTI Controversies:** Empirical limitations acknowledged

## Future Directions

1

**Personality Accuracy:** Improve prediction methods

2

**Global Expansion:** More countries and languages

3

**Open-Source Models:** Explore larger context models

4

**Alternative Frameworks:** Big Five integration

## Methodological Note

MBTI chosen for computational tractability and hierarchical cognitive process modeling, despite empirical limitations compared to Big Five

# Conclusion

## Key Achievements

- **Successful Integration:** Psychology theory enhances AI survey simulation
- **Significant Improvements:** 10% accuracy gains with better interpretability
- **Cross-Cultural Validation:** Robust performance across Eastern and Western cultures
- **Interdisciplinary Bridge:** Opens new research directions at AI-psychology intersection

## Research Impact

MARK demonstrates the potential of **psychology-informed AI** to advance computational social science and enable cost-effective, scalable human behavior modeling.

The future of survey research is computational, interpretable, and grounded in human psychology.



Ontoweb  
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MARK code  
Github Link



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# THANKS

## DO YOU HAVE ANY QUESTIONS?

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**Code Repository:**

<https://github.com/alexc-l/MARK-Multi-stAge-Reasoning-framework>

**Institution:** Wuhan University of Science and Technology

Questions & Discussion Welcome!

Slides available at: EMNLP 2025

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